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QUIZ 1

Given that two functions (**funA** and **funB**) have been defined as follows:

```cpp
void funA(int n) {
  cout << "funA - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

```cpp
void funB(int n) {
  cout << "funB - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

Study the funA and funB functions above. Identify the followings:

1. Instruction to check if the terminal case has been reached: ______if(n<5)______________________

2. Instruction to change the size of the problem so the terminal case can be reached:
____n=n+1__________________

Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```cpp
int main() {
  funA(1);
  return 0;
}
```

```
void funA(1) {
        cout<< "funA - " << 1 << "\n";
        n=1+1;
        if (2<5) {
                if(2%2==0) {
                        funB(2);
                } else{
                        funA(n);
                }
        }
}
```

Output:
funA - 1

```
void funB(2) {
        cout<< "funB - " << 2 << "\n";
        n=2+1;
        if (3<5) {
                if(3%2==0) {
                        funB(n);
                } else{
                        funA(3);
                }
        }
}
```

Output:
funA - 1
funB - 2

```
void funA(3) {
        cout<< "funA - " << 3 << "\n";
        n=3+1;
        if (4<5) {
                if(4%2==0) {
                        funB(4);
                } else{
                        funA(n);
                }
        }
}
```

Output:
funA - 1
funB - 2
funA - 3

Reach the terminal case

```
void funB(4) {
        cout<< "funB - " << 4 << "\n";
        n=4+1;
        if (5<5) {
                if(3%2==0) {
                        funB(n);
                } else{
                        funA(3);
                }
        }
}
```

Output:
funA - 1
funB - 2
funA - 3
funB - 4

```
void funA(1) {
	cout<<”funA - “<<1<<”\n”;
	n=2;
	funB(2);
}
```

Output:
funA - 1

```
void funB(2) {
	cout<<”funB - “<<2<<”\n”;
	n=3;
	funA(3);
}
```

Output:
funA - 1
funB - 2

```
void funA(3) {
	cout<<”funA - “<<3<<”\n”;
	n=4;
	funB(4);
}
```

Output:
funA - 1
funB - 2
funA - 3

```
void funB(4) {
	cout<<”funB - “<<4<<”\n”;
	n=5;
	//reached terminal case
}
```

Output:
funA - 1
funB - 2
funA - 3
funB - 4